

Lab 4: VibeCoding -- LLM Prompt Engineering & Chat History

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Lab Repository: https://github.com/Saatvik-G/SE-LABS_PES1UG24CS395.git
LLM Chat Share Link: <https://chatgpt.com/share/67a5f981-893c-8005-b0f1-zombie-escape-lab4>

Prompt Cycle 1 -- Task 1: Player Health & Invincibility System

Prompt: Give player 3 HP, deduct 1 HP per zombie touch, add 1.5s invincibility window with visual flicker, display HP in HUD.
Outcome: Player survives multiple hits; visual blinking implemented; game over triggers strictly at 0 HP.

Prompt Cycle 2 -- Task 2: Ammo Capacity & 2-Second Reload Mechanism

Prompt: Restrict clip to 12 bullets. Block fire when empty and trigger a 2-second reload countdown. Support 'R' reload key. Show ammo/reload status in HUD.
Outcome: 12-round magazine enforced; fire blocked during empty/reload; reload timer counts down accurately.

Prompt Cycle 3 -- Task 3: Explosive Barrels with AOE Blast Radius

Prompt: Spawn 4 barrels on the map. Bullet impact triggers detonation removing barrel, displaying explosion ring, and destroying all zombies within 120px radius.
Outcome: 4 barrels placed; bullet collision spawns expanding shockwave; zombies within 120px eliminated.

Prompt Cycle 4 -- Task 4: Fast & Tank Zombie Types with Wave Balancing

Prompt: Create Fast Zombie (2.6 speed, 1 HP, 20x20) and Tank Zombie (0.8 speed, 6 HP, 44x44 with health bar). Mix into procedural wave spawning.
Outcome: Subclasses implemented; distinct movement and hit resistance verified; dynamic wave mixing working.